Jia (家) Framework

AI-Native Cyber Defense for the Philippine Judiciary

Protecting the Digital Foundations of Justice

Supreme Court • Court of Appeals • RTC • MeTC • MTC • MCTC

The Challenge

Philippine Courts Are Under Digital Siege

The judiciary's digital transformation — eCourt, eFiling, virtual hearings — has created an **unprecedented attack surface**.

Courts are uniquely high-value targets:

- 📁 **Sealed case records** — criminal, family, CICL, witness protection
- 🪪 **Sensitive PII** — litigants, minors, abuse victims, informants
- ⚖️ **Judicial integrity** — orders, decisions, deliberations must be tamper-proof
- 💰 **Financial records** — bail, fines, estate proceedings
- 🔗 **Evidence chains** — digital evidence must remain forensically intact

The Threat Landscape

What Philippine Courts Face Today

Threat Vector	Real-World Risk
SQL Injection	Attacker modifies case records directly in the eCourt database
Credential Theft	Stolen judge/clerk passwords → fraudulent court orders
Ransomware	Court operations halted; sealed records held hostage
Insider Threats	Unauthorized access to sealed family/CICL records
Malicious eFiling	Webshells disguised as court document uploads
DDoS Attacks	Filing portals disrupted during critical deadlines
Quantum Harvest	Encrypted records captured today, decrypted by future quantum computers

The cost of inaction:

Introducing Jia (家)

The Next-Generation AI Cyber Defense Platform

Jia (家) — meaning *"home"* in Chinese — protects your digital house.

A dual-engine defense system purpose-built for critical infrastructure:

Engine	Technology	Role
OTP Orchestrator	Gleam (Erlang/BEAM VM)	Fault-tolerant supervision, clustering, zero-downtime recovery
Vella SecOps Engine	Rust (Axum)	AI threat detection, kernel inspection, post-quantum cryptography

Key differentiators:

Performance Advantage

Jia vs. Legacy Security Solutions

Metric	🛡️ Jia	Legacy WAF	Python/JS Proxy
P99 Latency	**1.18 ms**	18.40 ms	45.20 ms
Throughput	**148,500 req/s**	22,100 req/s	6,400 req/s
Crash Recovery	**< 0.001 ms**	120 ms	850 ms
Memory	**24 MB**	180 MB	310 MB
PQC Signing	**0.34 ms**	❌ N/A	❌ N/A

> Jia processes an eFiling submission's security check in **under 2 milliseconds** — lawyers and clerks experience zero delay.

Why this matters for courts:

Defense in Depth

The 4-Stage Security Pipeline

Every request passes through **four independent defense layers** before reaching court systems:

Stage 1: YARA Signature Scanning

Detects known malware — shellcode, PHP webshells, SQL injection exfiltration

Stage 2: AI Firewall & PII Guard

Regex-based PII redaction + LLM-as-a-Judge prompt safety classification

Stage 3: AI Risk Assessment Engine

PII Protection & Data Privacy

Automatic Compliance with RA 10173

Jia's PII scrubbing engine **automatically detects and redacts** sensitive information:

Data Type	Example	Action
Social Security / Government IDs	`12-3456789-0`	→ `[REDACTED_PII]`
Credit Card Numbers	`4111-1111-1111-1111`	→ `[REDACTED_PII]`
API Keys & Secrets	`AKIA...`, `ghp_...`, `sk_...`	→ `[REDACTED_PII]`
Email Addresses	`judge@court.gov.ph`	→ `[REDACTED_PII]`
Prompt Injection Attacks	`"ignore instructions"`	→ `[FILTERED]`

Critical for Philippine courts:

The Digital Notary

Immutable WORM Audit Chain

Jia maintains a **Write-Once-Read-Many (WORM)** audit ledger — a hash-chained, cryptographically signed record of every security event:

```
GENESIS → Entry 1 (SHA-256) → Entry 2 (SHA-256) → Entry 3 (SHA-256) → ...
                                                   ↓
                                         ML-DSA-65 Quantum Signature
```

Each entry records:

- **What** happened (quarantine, block, alert, rollback)
- **When** it happened (RFC 3339 timestamp)
- **Who/What** was involved (IP, target, user)

Post-Quantum Cryptography

Protecting Court Records for Decades

The "Harvest Now, Decrypt Later" Threat:

Adversaries can **capture encrypted court data today** and decrypt it when quantum computers become available (estimated 2030–2040).

A sealed family court record from **2025** must remain sealed in **2050**.

Jia implements NIST-approved quantum-resistant algorithms:

Algorithm	Standard	Purpose
ML-KEM-768	FIPS 203 (Kyber)	Quantum-safe key exchange for secure channels
ML-DSA-65	FIPS 204 (Dilithium)	Quantum-safe digital signatures on audit logs

Passwordless Judicial Authentication

FIDO2 / WebAuthn Hardware Security

The problem with passwords:

- Phished via fake login pages targeting court staff
- Shared among clerks in busy municipal courts
- Reused across personal and court accounts

Jia's solution — eliminate passwords entirely:

- 🔑 **Hardware security keys** (YubiKey) for judges and senior staff

eCourt & eFiling Protection

Shielding Internet-Facing Court Systems

Jia deploys as a **security sidecar** in front of eCourt application servers:

```
Lawyer's eFiling → Jia Sidecar (< 2ms) → eCourt Server → Case Database
                        ↓ (if malicious)
                   BLOCKED + WORM Log + Alert
```

Protection against:

- ✅ SQL injection targeting case databases
- ✅ XSS in uploaded court documents
- ✅ Webshells disguised as PDF filings
- ✅ DDoS floods during filing deadlines (300 req/sec/IP limit)

Threat Intelligence & Response

MITRE ATT&CK + Automated SOAR

Real-time threat matching:

Jia's RAG engine performs **vector similarity search** against known attack patterns:

- Log4Shell (CVE-2021-44228)
- ProxyLogon (CVE-2021-26855)
- MOVEit SQLi (CVE-2023-34362)
- Spring4Shell (CVE-2022-22965)
- LLM Prompt Injection patterns

Kernel-Level Protection

eBPF System Call Monitoring

Jia monitors the **operating system kernel itself** via eBPF — catching threats that network-level tools completely miss:

Monitored syscalls:

- `execve` — unauthorized program execution
- `ptrace` — memory inspection / debugging (rootkit technique)
- `bpf_cmd` — attempts to tamper with eBPF monitoring itself

Detection targets:

- 🔍 Rootkit installation attempts

Regulatory Compliance

Alignment with Philippine Law

Law / Regulation	Requirement	Jia Feature
RA 10173 (Data Privacy Act)	Protect personal information	PII scrubbing, access logging, WORM audit
RA 10175 (Cybercrime Prevention)	Digital evidence integrity	SHA-256 hash chains, ML-DSA-65 signatures
A.M. No. 01-7-01-SC (Electronic Evidence)	Digital evidence admissibility	Cryptographic integrity proofs, ZK sharing
A.M. No. 12-12-11-SC (CICL Rules)	Confidentiality of minor's records	Automated PII redaction
NCSP 2023-2028	Critical infrastructure protection	eBPF kernel monitoring, MITRE ATT&CK intelligence

Self-Testing & Continuous Verification

Built-in Purple Team Simulation

Jia can **attack itself** to continuously verify defensive readiness:

5 automated attack vectors:

1. 💉 **Prompt Injection** — tests AI guardrail response
2. 🗃️ **SQL Injection** — tests database protection
3. 🍯 **Honeypot Violation** — tests trap detection
4. 🌊 **eBPF Syscall Flood** — tests kernel-level defense
5. ☠️ **RAG Poisoning** — tests document sanitization

Real-Time Command Dashboard

Glassmorphism Cyber Command Center

Jia includes an embedded, zero-dependency web dashboard at `http://localhost:9090/dashboard`:

Dashboard capabilities:

- 📊 **Live Threat Feed** — real-time WebSocket telemetry stream
- 💻 **Node Cluster Status** — BEAM + Rust sidecar health
- 📜 **WORM Audit Inspector** — searchable, signed audit ledger
- 🔍 **RAG CVE Search** — interactive MITRE ATT&CK lookup
- 🧹 **PII Scrubber Lab** — test PII redaction in real-time

Deployment Strategy

Tiered Rollout Across the Court Hierarchy

Court Level	Deployment Model	Priority
Supreme Court	Full dual-engine, dedicated hardware, eBPF, PQC, central SIEM	Phase 1
Court of Appeals	Full deployment, cluster gossip to SC SOC	Phase 1
RTC (Regional)	Sidecar alongside eCourt infrastructure	Phase 2
MeTC (Metropolitan)	Lightweight Rust engine, centrally managed	Phase 3
MTC / MCTC (Municipal)	Cloud-hosted shared Jia instance	Phase 3

Why this works at every level:

- **24MB memory** — runs on existing municipal court hardware
- **Sub-microsecond crash recovery** — stays online on unreliable infrastructure

Pilot Program Proposal

Recommended First Steps

Phase 1 — Proof of Concept (3 months)

1. **Deploy** Jia sidecar alongside the Supreme Court eCourt system
2. **Monitor** — observe threat detection, PII redaction, and audit logging in passive mode
3. **Validate** — run purple team simulation, verify WORM chain integrity
4. **Report** — present findings to the Committee on Information Technology

Phase 2 — Expansion (6 months)

4. **Activate** blocking mode at Supreme Court

Why Jia

The Case for Open-Source Judicial Cyber Defense

Advantage	Impact
🔒 Immutable audit trail	Proves court records haven't been tampered with
⚛️ Post-quantum cryptography	Protects sealed records for decades
🪪 Automated PII protection	Enforces RA 10173 at infrastructure level
⚡ Sub-2ms latency	Zero impact on court operations
🚫 Fail-closed security	Blocks threats even when AI services are offline
🔑 Passwordless auth	Eliminates credential theft as an attack vector
🧪 Self-testing capability	Continuous verification without external auditors
💾 24MB footprint	Deployable from SC data centers to MTC offices

Jia (家)

Protecting the Digital Foundations of Philippine Justice

Open Source • Quantum-Ready • Built for Courts

Contact: github.com/CharleGutierrez/jia